

Geonmo Gu

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🌐 <https://github.com/gmgu>

RESEARCH INTERESTS

Humanoid AI: Vision-Language-Action Model, Progress Reward Model based on Vision-Language Model, Open-Loop Evaluation using Simulator.

Large Language Model: AI Coding Assistant, Code LLM, Multi-Node Distributed Training, Parameter Efficient Fine-Tuning, Instruction Tuning, LLM Inference Server, Prompt Engineering, Benchmark Dataset, Data Collection and Cleaning, Time Series Forecasting, Semantic Parsing.

Algorithm Engineering: Fast and Scalable Algorithms, Graph Isomorphism, Subgraph Matching, Multiple String Matching, Cartesian Tree Matching, Order-Preserving Matching, Traveling Salesman Problem, Approximating Polygons and Subdivisions with Minimum-Link Paths, Path Simplification.

WORK EXPERIENCE

LG Electronics

Seoul, Korea

Principal Researcher

Mar. 2026 – Present

Artificial Intelligence Lab, Physical Intelligence Task

- Humanoid AI (Mar. 2026 – Present)

Senior Researcher

Apr. 2022 – Feb. 2026

Artificial Intelligence Lab, AI Algorithm TP

- Humanoid AI (Jan. 2025 – Feb. 2026)
- Development of AI Coding Assistant using Large Language Model (Jun. 2022 – Dec. 2024)
- Development of Coding Education Program Utilizing AI (Apr. 2022 – Dec. 2022)

Seoul National University

Seoul, Korea

Postdoctoral Research Assistant

Sept. 2021 and Jan. 2022 – Mar. 2022

Institute of Computer Technology

- Algorithm Development for Graph Isomorphism Query Processing

NAVER

Gyeonggi-do, Korea

Internship

Oct. 2021

AI Dev2

- Analyzing Conversion Tracking Data

EDUCATION

Seoul National University

Seoul, Korea

Ph.D. in Computer Science and Engineering

Mar. 2014 – Aug. 2021

- Thesis: Fast Graph Isomorphism using Pairwise Color Refinement and Efficient Backtracking
- Advisor: Prof. Kunsoo Park
- GPA: 3.99/4.3

Incheon National University

Incheon, Korea

B.S. in Computer Science and Engineering

Mar. 2010 – Feb. 2014

- GPA: 4.4/4.5 (summa cum laude)

PUBLICATIONS

Geonmo Gu, Jaeho Kwak, Haksoo Moon, Hyun Seung Shim, Yu Jin Kim, Byoungjip Kim, Moontae Lee, Hyejeong Jeon. "Overlapping Context with Variable-Length Stride Increases Diversity when Training Large Language Model for Code." *Annual Meeting of the Association for Computational Linguistics (Industry)*, 2025.

- Developed a training-data processing method for code LLM to mitigate the data-scarcity issue.

Geonmo Gu, Yehyun Nam, Kunsoo Park, Zvi Galil, Giuseppe F. Italiano, and Wook-Shin Han. "Efficient Graph Isomorphism Query Processing using Degree Sequences and Color-Label Distributions." *IEEE International Conference on Data Engineering*, 2022.

- Developed a fast algorithm for graph isomorphism query processing.
- Graph isomorphism query processing can be applied to chemistry database search.

Geonmo Gu, Yehyun Nam, Kunsoo Park, Zvi Galil, Giuseppe F. Italiano, and Wook-Shin Han. "Scalable Graph Isomorphism: Combining Pairwise Color Refinement and Backtracking via Compressed Candidate Space." *IEEE International Conference on Data Engineering*, 2021.

- Developed a fast and scalable algorithm for graph isomorphism.
- Graph isomorphism is a core problem in graph analysis of various domains, e.g., social network anonymization and circuit verification in VLSI design.

Siwoo Song, **Geonmo Gu**, Cheol Ryu, Simone Faro, Thierry Lecroq, and Kunsoo Park. "Fast Algorithms for Single and Multiple Pattern Cartesian Tree Matching." *Theoretical Computer Science*, 2020.

Geonmo Gu, Siwoo Song, Simone Faro, Thierry Lecroq, and Kunsoo Park. "Fast Multiple Pattern Cartesian Tree Matching." *International Conference and Workshop on Algorithms and Computation*, 2020.

- Developed a fast algorithm for multiple pattern Cartesian tree matching.
- Cartesian tree matching can be applied to time series data such as stock price analysis.

Myoungji Han, Hyunjoon Kim, **Geonmo Gu**, Kunsoo Park, and Wook-Shin Han. "Efficient Subgraph Matching: Harmonizing Dynamic Programming, Adaptive Matching Order, and Failing Set Together." *ACM SIGMOD International Conference on Management of Data*, 2019.

- Developed a fast algorithm for subgraph matching (number of citations: 210).
- Subgraph matching has a wide range of applications including RDF query processing, protein interaction analysis, chemical compound search, and social network analysis.

Myoungji Han, Munseong Kang, Sukhyeun Cho, **Geonmo Gu**, Jeong Seop Sim, and Kunsoo Park. "Fast Multiple Order-Preserving Matching Algorithms." *International Workshop on Combinatorial Algorithms*, 2015.

Seongi Hong, **Geonmo Gu**, Hyunjoon Kim, Kunsoo Park. "Performance Comparison of Adaptive Matching Orders for the Subgraph Isomorphism Problem." *KIISE Transactions on Computing Practices*, 26.1:38-43. 2020.

Seongi Hong, **Geonmo Gu**, Hyunjoon Kim, Kunsoo Park. "Performance Comparison of Candidate-Size Ordering and Path-Size Ordering for Subgraph Isomorphism Problem." *Korea Computer Congress*, 2019

PROJECTS

Development of AI Coding Assistant using Large Language Model

LG Electronics

AI Algorithm TP

Jun. 2022 – Dec. 2024

- User: employees in LG Electronics.
- Service: code suggestions for webOS and general programming.
- Architecture: VS Code extension, VIM plugin, billion-scale LLMs, vLLM server on DGX GPUs.

Development of Coding Education Program Utilizing AI

LG Electronics

AIX Lab

Apr. 2022 – Dec. 2022

- User: middle and high school students who want to learn Python.
- Service: generating Python code from natural language instruction.
- Architecture: web client, an encoder-decoder transformer, inference server on RTX GPUs.

Analyzing Conversion Tracking Data

NAVER

AI Dev2

Oct. 2021

- Conducted exploratory data analysis on glad for advertisement data to find meaningful trends.
- Handled hundred gigabytes of (raw) conversion tracking data.
- Solved optimization problem of maximizing conversion rate using linear programming.

Framework of Practical Algorithms for NP-hard Graph Problems

Seoul National University

Funded by the Korea government (Ministry of Science and ICT)

Apr. 2018 – Aug. 2021

- Algorithm development for fast subgraph isomorphism, graph isomorphism, and graph isomorphism query processing.
- Open source contribution for practical graph algorithms (<https://github.com/SNUCSE-CTA>).

Algorithm Development for Scanner/Stage Path Generation

Seoul National University

Supported by JASTECH

Jul. 2014 – Jun. 2017

- Sophisticated algorithm that can synchronize Scanner and Stage.
- Development of path simplification method based on chain stabbing (computational geometry).
- Efficient path generation methods by solving the traveling salesman problem (NP-complete).

NIPA-PURDUE Capstone Program

Purdue University

Center for Robotic Innovation, Commercialization and Education

Jan. 2014 – Feb. 2014

- Robot programming (Robotis Bioloid) in collaboration with students of Purdue University.

PROFESSIONAL ACTIVITIES

Reviewer of IEEE Access

IEEE

Sep. 2025 – May. 2026

Seminar about Beyond Behavior Cloning VLA Techniques

LG Electronics

Nov. 2025

Seminar about Distributed Training Large Language Models

Dankook University

Apr. 2024

Reviewer of Information Processing Letters

ELSEVIER

Dec. 2020 – Sep. 2023

Seminar about Distributed Training Techniques for Large AI Models

LG Electronics

Jul. 2023

Invited talk at STARLAB Meeting

Korea Computer Congress

Jun. 2023

Invited talk at 2023 TOPCIT Workshop

IITP
Mar. 2023

HONORS

2023 Innovation Awards of CTO Division

Awarded a Grand Prize

LG Electronics

Jan. 2024

The 14th Open SW Developer Contest

Awarded a Gold Prize

Ministry of Science and ICT

Nov. 2020

The 2nd Test of Practical Competency in IT (TOPCIT)

Awarded a Silver Prize

Ministry of Science and ICT

Sep. 2013

The 1st Test of Practical Competency in IT (TOPCIT)

Awarded a Grand Prize

Ministry of Knowledge Economy

Oct. 2012

SKILLS

Programming Languages. C/C++, Python, CUDA C++, Rust, C#, Java, Shell Script, L^AT_EX

- C++: <https://github.com/gmgu/GI>
- CUDA C++: <https://github.com/gmgu/study-cuda>
- Rust: <https://github.com/gmgu/study-rust>

Libraries. PyTorch, TensorFlow, HuggingFace Transformers, DeepSpeed, Triton (NVIDIA), Faster-Transformer, FastAPI, Triton (OpenAI), Seaborn, Pandas, PySpark, gtest

- Triton: <https://github.com/gmgu/study-triton>

Competitive Programming.

- BAEKJOON: <https://www.acmicpc.net/user/gmgu>

Framework. AWS (SageMaker, EC2, Lustre, S3), Docker, ROS 2